ABHIK RAO | 2025025

COMPUTER SCIENCE AND ARTIFICAL INTELLIGENCE (CSAI)

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2029	B.Tech in CS and Artificial Intelligence	IIIT Delhi	8.60 / 10
2025	Class XII, CBSE	New Heera Lal Public School	87%
2023	Class X, CBSE	RPS Public School	95.6%

HACKATHONS

Autonomous Job Application Agent |TinyFish $2M Pre-Accelerator Hackathon — Top 15/4,300+ Teams

- Ranked among the Top 15 teams out of 4,300+ registrations, advancing to the Phase2 Accelerator
- Designed an AI-powered autonomous agent to automate job applications across Greenhouse, Lever, Workday, and LinkedIn
- Leveraged LLM-assisted development to rapidly prototype, integrate, and present an autonomous job application agent during the online Demo Day

SafeSiteAI | Edge-First Construction Safety System |Qualcomm Snapdragon Multiverse Hackathon — Top 8 Finalist

- Achieved Top 8 finalist status among 60 shortlisted teams at Qualcomm's Snapdragon Multiverse Hackathon
- Developed a real-time construction safety system to detect PPE violations, restricted-zone entry, and falls using YOLO and OpenCV
- Integrated the web dashboard, Arduino-based alarm system, and edge-to-edge inference pipeline while debugging and presenting the final solution

PROJECTS

DSA Search Engine | Self-Project

Overview: A search engine for 3,500+ DSA problems using automated web scraping, TF-IDF, cosine similarity, and an Express.js backend API

- Compiled a 3,500+ DSA problem corpus via Puppeteer, automating dynamic scraping & structured extraction from LeetCode & Codeforces
- Enhanced search relevance with TF-IDF and cosine similarity, using preprocessing and title boosting to rank the top 10 matching problems
- Streamlined query serving with precomputed document vectors and an Express.js API for efficient similarity scoring and ranked retrieval

Medical Prescription Understanding Pipeline | Research Project under Dr. Vinish Kutharia, IIIT Delhi

- Engineered a medical prescription understanding pipeline using TrOCR, LayoutLMv3, BioBERT, OpenCV, and Gradio for handwritten medicine recognition
- Fine-tuned LayoutLMv3, integrated OCR with drug validation modules
- Achieved 84.6% precision on the MIRAGE benchmark through synthetic data augmentation

Rubik's Cube Solver | Self Project

Overview: A C++-based Rubik's Cube Solver leveraging object-oriented design and advanced search strategies like IDA, BFS, and DFS*

- Simulated a virtual Rubik's Cube (3x3) using C++ with efficient modeling via STL containers like unordered_map and modular class design
- Applied core OOP concepts such as virtual functions, inheritance, and operator overloading to significantly enhance code reusability
- Utilized the efficient Korf's IDA Algorithm* and reliably achieved a solving time of under 10 seconds for a Rubik's Cube jumbled 13 times
- Reached reliable solving time under 3 seconds for a Rubik's Cube jumbled 8 times using optimized BFS, DFS, and IDDFS algorithms

Environmental Sound Classification using AST | Self Project

- Developed an environmental sound classification model using a pretrained Audio Spectrogram Transformer (AST) on the ESC-50 dataset
- Implemented linear probing by freezing the pretrained encoder and training a lightweight 50-class classifier
- Achieved 93.25% test accuracy while fine-tuning only 0.046% of the model's 86M+ parameters

RISC-V Assembler & Simulator| Python

- Implemented a RISC-V assembler and simulator supporting R, I, S, B, U, and J instruction formats with virtual register and memory mangement
- Developed robust error handling for invalid instructions, syntax errors, and memory access violations
- Implemented virtual registers to simulate instruction execution and stage changes

SKILLS AND EXPERTISE

Languages & Libraries: SQL | Python | HTML | C/C++ | LaTeX | MATLAB | R | Java | NumPy | Pandas | Matplotlib | BeautifulSoup | Sklearn | Streamlit
Software & Tools: Power BI | QuickSight | Metabase | Google Analytics | MS Clarity | Figma | Advanced Excel | MS Office Suite | Jupyter Notebook | Git
Frameworks: PyTorch | OpenCV | React.js | Next.js | Tailwind CSS | Hugging Face | Gradio

COMPETITION/CONFERENCE

- Attained a peak rating of 1734 on LeetCode, placing among the top 20% of users on the platform
- Achieved a peak rating of 1499 (2★) on CodeChef under the handle bliss vista77
- Participated in competitive programming on Codeforces under the handle ovoabhik, attaining a peak rating of 1161
- Attained a prestigious rank of 2 on InterviewBit by successfully solving over 150+ coding problems

ACHIEVEMENTS

- Achieved a top 10 percentile rank among 1.8 lakh candidates in the JEE Advanced 2025 conducted by IIT Kanpur
- Secured a top 2 percentile rank among 10.6 lakh candidates in JEE Main 2025
- Secured an All India Rank within the Top 300 in the National Standard Examination in Astronomy (NSEA)

LEADERSHIP & CAMPUS INVOLVEMENT

Induction Mentor | IIIT Delhi

Mentored incoming first-year students during IIIT Delhi's induction programme, supporting their academic and campus transition.

Community Work

Volunteered with CanWinn Foundation for 80+ hours, supporting medical camps, patient guidance, beneficiary registration, health awareness drives, and community outreach